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Research article



Biochar benefits carbon off-setting in blue-green infrastructure soils - A lysimeter study

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ABSTRACT

Carbon sequestration with amendments in blue-green infrastructure soils could off-set anthropogenic greenhouse gas emissions to alleviate climate change. In this 3-year study, the effects of wheat straw and its biochar on carbon sequestration in an urban landscaping soil were investigated under realistic outdoor conditions using two large-scale lysimeters. Both amendments were carried out by incorporating pellets at 0–15 cm soil depth with an equivalent initial total carbon input of 2% of the dry soil weight. Soil carbon, carbon isotope ratios, dissolved carbon in leachates, CO_2–C emissions, carbon fixed in above ground vegetation, soil water content, soil bulk electrical conductivity, and water infiltration rates, were then compared between the 2 lysimeters. After 3 years, we observed that, despite having a 17.2% lower vegetation growth, soil organic and inorganic carbon content was higher by 28.8% and 41.5%, respectively, in biochar as compared to wheat straw amended soil. Carbon isotope analysis confirmed the greater stability of the added carbon in the biochar amended soil. Water content was on average 23.2% and 13.0% in the straw pellet and biochar amended soil, respectively, whereas water infiltration rates were not significantly different between the two lysimeters. Overall, the incorporation of wheat straw biochar into soil could store an estimated 30 tonnes of carbon per hectare in city blue-green infrastructure spaces. Interviews involving institution stakeholders examined the feasibility of this biochar application. Stakeholders recognized the potential of biochar as an environment-friendly means for carbon offsetting, but were concerned about the practicality of biochar production and application into soil and increased maintenance work. Consequently, additional potential benefits of biochar for environmental management such as improving the quality of polluted run-off in stormwater treatment systems should be emphasized to make biochar an attractive proposition in sustainable urban development.

1. Introduction

By December 2020, 1807 local governments (cities/towns) across 33 countries had declared a climate emergency (Cedamia, 2020) and introduced guidelines to regulate their carbon emissions for alleviating anthropogenic climate change. Many governments aim to achieve city-wide carbon neutrality in the coming years, e.g. Sydney by 2040 (City of Sydney News, 2020), and Greater London (Thorpe, 2020) and Newcastle City Council (2020) by 2030. Since fossil fuel usage remains the dominant source of anthropogenic CO_2 emissions, and complete replacement of fossil fuel usage is unlikely in the short-term, an approach to offset carbon emissions is inevitably required for carbon neutrality (El-Naggar et al., 2019). According to The Royal Society & Royal Academy of Engineering (2018), urban green space offers opportunity to offset city carbon emissions in two ways: a) biomass increase of vegetation and b) carbon sequestration in urban soils. Carbon sequestration is the capture, removal, and storage of CO_2 from the atmosphere into the other carbon storage reservoirs such as vegetation or soils. In urban settings, carbon sequestration using biochar as a soil amendment is attractive during construction projects because of additional benefits in blue-green city spaces. Biochar implementation in soil slowly releases diverse plant nutrients and stimulates geochemical cycling, whilst remediating the high compaction of urban soils to support plant growth and reduce water pollution problems (Das and Ghosh, 2021). Biochar is charcoal-like material produced from biomass by thermochemical conversion with limited oxygen, a process with generates pyrogenic carbon that can endure in soil for thousands of years (Lehmann and Joseph, 2015). Biochar properties depend on the

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feedstocks which include wood chips, agricultural residues, sewage sludge, dairy manure, etc, and also production conditions such as pyrolysis temperature and kiln residence time (Das and Ghosh, 2021). All these variables can affect the biochar properties and stability, and impact the function of biochar on soil carbon sequestration (Chatterjee et al., 2020; El-Naggar et al., 2018). Biochar produced from straw has shown various benefits in soil such as increased field capacity (Das and Ghosh, 2022), crop yield (Karhu et al., 2011), sorption of organic contaminants (Song et al., 2012), and creating a fertile soil micro-environment (Taherymoosavi et al., 2018). Moreover, utilizing local straw residues for biochar production can reduce haulage and valorise such waste, being a good example of a circular economy (Zhang et al., 2021).

Most previous studies of biochar focused on agricultural soil amendment benefits such as increasing soil nutrients, hydrological pathways and decomposition of organic matter (Jiang et al., 2022; Huang et al., 2018). Soil aggregate carbon and humic substances carbon were enhanced by approximately 30% in cropping soil in northeastern China after adding maize straw or biochar (Sun et al., 2022); Wang et al. (2021b) observed a gain of different categories of carbon and enzyme activities with the presence of wheat straw or straw biochar in a tobacco field, that resulted in a 10.6–92.2% increase of soil fertility. A robust soil aggregate cohesion and water retention effect occurred in seasonally frozen soils treated by both biochar or straw (Xue et al., 2022), while a greater carbon stability was found with the biochar treatment for indicators like soil respiration rates, but the authors did not compare the net change of total soil carbon mass between two practices. Additionally, Duan et al. (2020) assessed the potential of sustainable agriculture by comparing greenhouse gas emissions from farmland soils amended with wheat straw or its biochar and found that biochar amendment did not affect soil CH_4 oxidation, but decreased CO_2 and N_2O emissions. Notably, previous studies revealed that soil amendment with biochar does not always simply result in a corresponding soil carbon increase (Huang et al., 2018). This is because of "priming effects" on native soil organic carbon mineralization, and impacts on soil inorganic carbon formation via changes in soil pH, calcium concentration, water content, etc. (El-Naggar et al., 2019). Hence it is imperative to understand the effects of biochar versus alternative soil amendments, such as straw, for carbon sequestration under realistic outdoor conditions (Laird et al., 2017).

Few studies have considered biochar application opportunities in the context of urban redevelopment, where amendments may benefit soil carbon storage and improve vegetation productivity in landscaping soils with fragmented construction materials (Somerville et al., 2020). If incorporated into blue-green infrastructure soil, biochar may additionally offer passive treatment of stormwater in Sustainable Urban Drainage Systems (SuDS) (Sun et al., 2020; Borah, 2020; Mohanty et al., 2014). In this context we hypothesized that wheat straw biochar amendment could more lastingly sequester carbon in urban soil than amendment with the unaltered wheat straw biomass. Based on the literature reports summarized above we also expected that vegetation growth would be higher, and soil hydraulic conductivity would be improved in the biochar amended urban soil to achieve multiple benefits. The objectives of this study were thus: 1) to compare the impact of two distinct urban soil amendment strategies on soil carbon storage and hydrology over 3 years; 2) to evaluate the soil carbon sequestration potential which the preferable of the two strategies would have, if implemented for the whole blue-green space of an urban redevelopment; 3) to evaluate the practical feasibility of the proposed carbon off-setting strategy with stakeholder interviews.

2. Materials and methodology

2.1. Study site and experimental design

2.1.1. Study site

We used the UK's National Green Infrastructure Facility (NGIF), which was established in the centre of Newcastle upon Tyne, North East England, on the Newcastle Helix site (54°58′45.0″ N, 1°36′49.8″ W), and is managed by Newcastle University. Newcastle University and Newcastle City Council have both committed to achieve carbon neutrality by 2030 (Bradley, 2019). Newcastle City experiences a temperate oceanic climate with annual temperature and precipitation of 8.5 °C and 693 mm, respectively (MetOffice, Zang et al., 2022). This study used the NGIF to investigate carbon sequestration opportunities via the amendment of urban soil with pellets of wheat straw biomass or its biochar, respectively.

2.1.2. Lysimeter set-up

Two identical lysimeters of the NGIF were used in this study (Fig. 1a). Each lysimeter was 4.5 m long, 2 m wide, and 1.1 m deep. The lysimeters were filled in layers with 80 cm of sand (<4 mm) on top of gravel in the conical lysimeter bottoms. A 15 cm thick topsoil layer (8% clay, 15% silt, 40% sand, and 37% coarse fragments >2 mm) was placed above the sand. The topsoil was from a local commercial provider of soil for landscaping, and contained fragments of construction materials, such as pieces of brick, slate roof tiles, and tarmac, in addition to clay, silt, and sand. The engineered landscaping soil texture, composition and chemical analysis can be found in Table S1-3 in Supporting **Information (SI)**. Sensors (ECH2O 5 TE, Meter Group, Weiz, Austria) were installed in both lysimeters to monitor soil temperature, volumetric water content, and bulk electrical conductivity, every 15 min, at 10 cm, 50 cm, and 80 cm depth.

The experimental design is outlined in Fig. 1b. Two topsoil amendment materials, wheat straw pellets (6 mm diameter), and biochar produced from these wheat straw pellets, were purchased from the UK Biochar Research Centre in Edinburgh, UK. The biochar was produced via pyrolysis with a residence of 15 min in a kiln at 700 °C (UK Biochar Research Centre, 2014) (Table S5). In June 2018, this biochar was added to one lysimeter at 4.8 kg m^{-2}, thereby aiming to achieve a 2% w/w total carbon augmentation of the 15 cm deep topsoil layer. The amendment summed up to a total of 43.20 kg of biochar (Table S4). Wheat straw pellets were applied to the other lysimeter at a higher rate aiming to achieve an equivalent total carbon weight amendment as the wheat straw biochar. Thus, the amount of wheat straw pellets added to the lysimeter was 7.15 kg m^{-2}, giving a total of 64.35 kg of wheat straw biomass pellets in that lysimeter. Pasture seeds, a mixture of ryegrass and white clover as used by the Newcastle University managed research farms, were applied over the two lysimeters.

2.2. Research methods

2.2.1. Soil carbon measurement and infiltration tests

The soil carbon monitoring was performed from June 2018 to June 2021, with a reduced monitoring schedule from April 2020 because of the Covid-19 pandemic. Before adding the amendments to the lysimeters, samples were collected randomly from the topsoil (0–15 cm) and the sand layer (40 cm). After the amendments, 0–15 cm soil samples were collected with a hand auger (UMS gouge auger, 25 mm diameter) within the four quadrants in each lysimeter, every three months in 2018, and every two months in 2019, plus twice a year in 2020 and 2021. Soil samples from 15 to 40 cm depth were collected once in 2018, three times in 2019, and twice a year in 2020 and 2021. Soil collection ≥40 cm depth was not considered, as changes in soil carbon were anticipated to occur mainly near where the amendments had been placed. Soil samples were prepared for carbon analysis using a LECO-RC 612 machine (LECO Corporation, Saint Joseph, Michigan, USA). A subset of 34 lysimeter soil

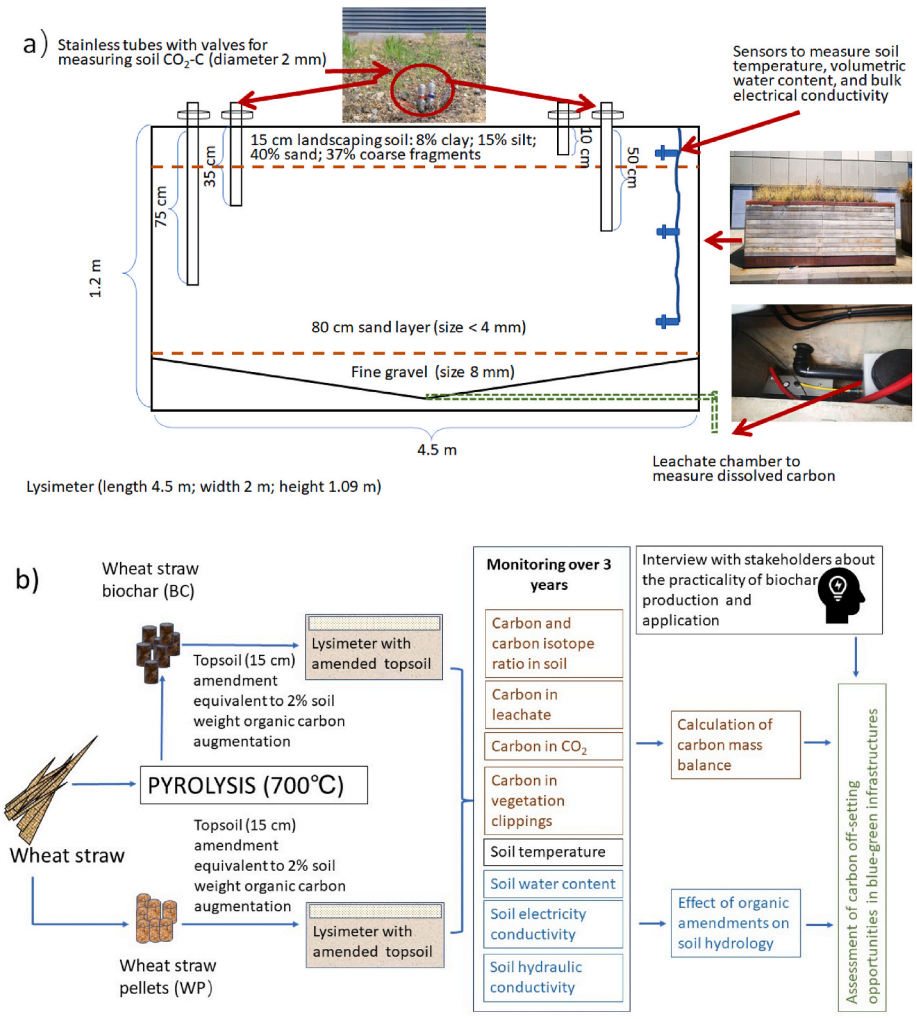


Fig. 1. Schematic illustrations of a) the lysimeter and b) the study design.

samples were selected for stable isotope analysis (20 topsoil and 14 subsurface samples) by Iso-Analytical Ltd (http://www.iso-analytical.co.uk). To understand amendment impacts on soil hydrology, double ring infiltration tests were carried out on the surface of the lysimeters monthly from February 2021–June 2021 (Johnson, 1963). The infiltration test was conducted in duplicate in each lysimeter on each occasion. The relevant methodologies pertaining to soil carbon measurement and calculation, carbon isotope analysis, use of carbon isotope data to establish sources of carbon in soils, and infiltration test results are provided in **SI**.

2.2.2. *Vegetation carbon measurement*

The biomass of vegetation in the two lysimeters from about 3 cm above the ground was determined by clipping in June 2019, August 2020, and June 2021. After weighing, the fresh clippings (apart from a small sub-portion used for the water and carbon content determinations) were reapplied onto the respective lysimeter surfaces, to mimic blue-green infrastructure management practices. Calculation of vegetation carbon content is given as Eq. S5 in **SI**.

2.2.3. *Leachate carbon measurement*

A drainage pipe (the green dash line in Fig. 1a) under each lysimeter collected leachate into a polypropylene chamber. One litre of leachate was collected from the chamber of each lysimeter biweekly (August 2018–February 2020) and stored in a cold room (4 °C) for pH and alkalinity measurements. For determining dissolved carbon (total & organic), leachate samples were filtered (0.45 μm, 25 mm; VWR International, UK) to separate dissolved from particulate carbon, and afterwards the dissolved carbon concentrations were measured by a carbon analyzer (Vario TOC cube, Elementar Analysen Systeme GmbH, Germany). The carbon loss from leaching was calculated by multiplying the concentration of dissolved carbon in leachate with the total rainfall volume falling on each lysimeter (minus evaporation) between two sampling dates referenced from World Weather Online (2018–2020). More details about the calculation and leachate composition measurements are provided in Eqs. S6-7 of **SI**.

2.2.4. *CO_2–C emission carbon measurement*

The CO_2 emissions were analysed monthly (August 2018–February 2020) by gas chromatography-mass spectrometry (Fisons 8060 GC/MD800MS). Four stainless steel tubes (2 mm diameter) in different lengths (10 cm, 35 cm, 50 cm, 75 cm) were sealed at the top with tube fittings and septa, and put in each lysimeter (Fig. 1a) so that gas samples from four soil depth layers could be obtained. From each tube, gas samples were obtained in triplicate. Meanwhile, air samples at 20 cm above the surface soil were also collected three times for measuring the CO_2 concentration in air above the lysimeters. More detailed steps in terms of gas collection and conversion of the CO_2 concentration ($g \cdot m^{-3}$) to CO_2–C fluxes ($g \cdot m^{-2} \cdot h^{-1}$) as well as the total emitted carbon mass are provided in Eqs. S9-14 in **SI**. At the end, the hourly CO_2–C flux at topsoil surface was multiplied with the total hours per month and the surface area of the lysimeter to estimate the monthly cumulative carbon lost

from the topsoil layer to the atmosphere. In addition, a portable EGM-4 Gas Monitor (PP Systems, Amesbury, USA) was used to compare the variation of CO_2–C fluxes between two methods, using Eq. S15-18 as explained in **SI**.

2.3. Carbon mass balance for the lysimeters

A carbon mass balance for each lysimeter considered the amount of carbon lost via leachate and soil CO_2–C emissions to the atmosphere, and the atmospheric CO_2–C captured in the above-ground vegetation and soil carbon accumulation. We assumed that carbon emitted during the biochar production is neutral because these emissions were counterbalanced with the carbon fixed from the atmosphere during the plant growth (Thakkar et al., 2016; Tisserant, 2019). In theory, the pyrolysis syngas could also be used for the generation of power to reduce fossil fuel usage, and the avoided carbon emissions in the pyrolysis process are then overall generally positive (Roberts et al., 2010; Thakkar et al., 2016).

2.4. Interview with estate and sustainability stakeholders

In the context of Newcastle University's aim to achieve net-zero carbon by July 2030, we gathered the expert opinions from Newcastle University Estate and Sustainability stakeholders on biochar application for soil carbon augmentation across the campus. We conducted structured qualitative interviews with a set of open-ended questions in December 2020 (Krull, 2018; Pensulo, 2012; Salo, 2018). Questions considered implications from the feedstock sourcing and biochar production stage to the practical deployment stage. Three key stakeholders comprising farm management, estate management, and institutional carbon management, were selected based on their role towards implementing net-zero carbon strategies for Newcastle University. The interviews notes are included in **SI**.

2.5. Statistical analysis

We performed statistical evaluation of the data sets with SPSS software (26.0, IBM Corp, Armonk, New York, USA). For the comparison of carbon values between the two systems we used One-Way ANOVA. If additional controlling factors were involved, such as soil temperature, soil depth or water content, we used Post Hoc tests (Tukey's HSD) for multiple comparisons between groups. If one group had more than two pairs of datasets (e.g. four soil depths in two lysimeters would have eight pairs of datasets in total), Fisher's Least Significant Difference (LSD) in Post Hoc tests was selected for comparing the pairs of data within the same group. We considered statistical significance at the $p \leq 0.05$ level.

3. Results

3.1. Soil carbon

Compared with the topsoil total carbon (TC) before the application of amendments (June 2018) (Fig. 2), the topsoil TC in the last collection (June 2021) had increased by 29.7% from 9.77 ± 1.46 kg m^{-2} to 12.68 ± 2 kg m^{-2} in the lysimeter amended with wheat straw biochar (lysimeter BC). In contrast, TC decreased by 3.71% from 9.77 ± 1.46 kg m^{-2} down to 9.41 ± 1.25 kg m^{-2} in the lysimeter amended with an equivalent amount of organic carbon using wheat straw biomass pellets (lysimeter WP). TC, total organic carbon (TOC), and total inorganic carbon (TIC), in lysimeter BC were all greater than lysimeter WP, throughout the experiment, although a significant statistical relationship was absent in TIC (Table S7). The final TOC of topsoil was 40.0% and 8.70% higher in BC and WP lysimeters, respectively, compared with the value measured before the amendment. Below the topsoil at around 15–40 cm depth, neither form of soil carbon was significantly affected by the two different amendments (TC: $p = 0.933$; TOC: $p = 0.689$; TIC: $p = 0.751$, Table S7). Furthermore, the variation analysis indicated that the TC contents in lysimeter BC, and TOC in both practices, were significantly different between the two soil depths ($p < 0.05$, Table S8).

In terms of isotopic composition, $\delta^{13}C$ in organic matter ($\delta^{13}C_{org}$) was

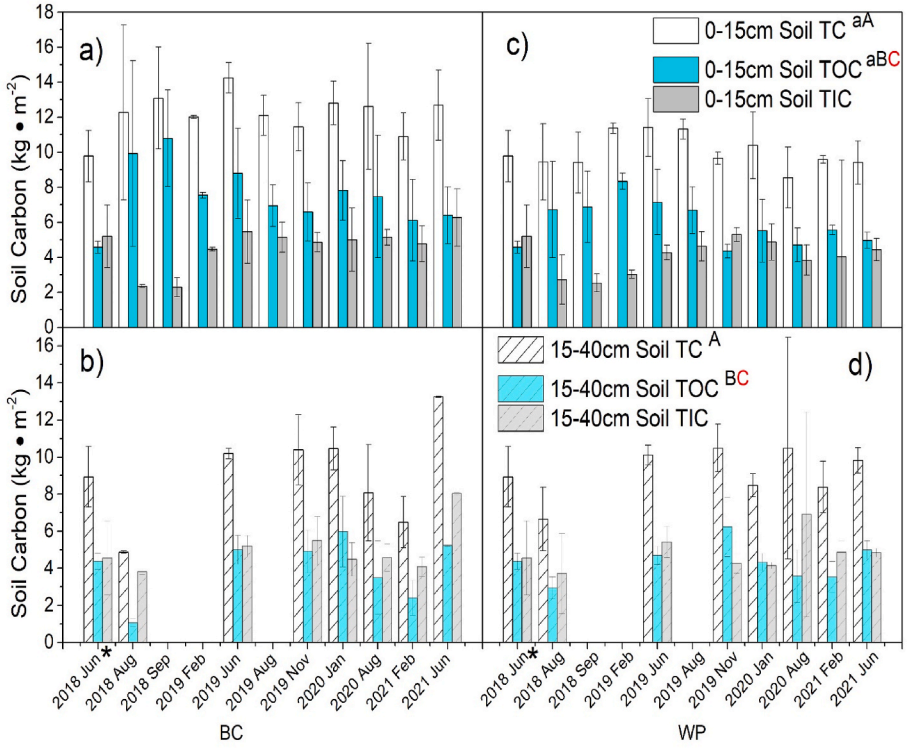


Fig. 2. Carbon storage (kg·m^{-2}, Mean ± Standard deviation) of soils in two lysimeters (June 2018–June 2021), *: the soil measurement in June 2018 is before the amendments; TC: total carbon; TOC: total organic carbon; TIC: total inorganic carbon; BC: lysimeter with wheat straw biochar; WP: lysimeter with wheat straw biomass pellets. a) soil carbon storage of topsoil (0–15 cm) in BC; b) soil carbon storage at 15–40 cm soil in BC; c) soil carbon storage of topsoil (0–15 cm) in WP; d) soil carbon storage at 15–40 cm soil in WP. The items with same superscript lowercase letters indicate significant differences between 2 treatments at $p \leq 0.05$ level; the items with same superscript capital letters in black indicate significant differences between two soil depths in BC ($p \leq 0.05$ level); the items with same superscript capital letters in red indicate significant differences between two soil depths in WP ($p \leq 0.05$ level). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

the lowest for pure wheat straw pellets and the biochar: −28.95‰ and −29.53‰ V-PDB, respectively. Consequently, $\delta^{13}C_{org}$ of the amended topsoil had a lower value throughout the experiment when compared with the topsoil before treatments (Fig. 3). The soil amendments induced a sharp decrease of $\delta^{13}C_{org}$ at both soil depths which was more notable in the lysimeter BC than WP. After 1 year, $\delta^{13}C_{org}$ of topsoil from both environments increased steadily until the end of the experiment, when the value in the lysimeter BC (−24.81‰) remained lower compared to WP (−24.46‰) ($p = 0.092$, Table S7). Both remained below the non-amended topsoil (−23.94‰). These fluctuations occurred in the 15–40 cm soil as well, but after three years the application of amendments led to a slightly more positive soil $\delta^{13}C_{org}$ (BC: −24.13‰; WP: −23.93‰) compared with the soil before amendment (−24.55‰) ($p = 0.025$, Table S7). According to the $\delta^{13}C_{org}$ for the topsoil layer, the average fraction of TOC derived from additives was almost 33.7% greater in lysimeter BC than WP (Table S9). Most $\delta^{13}C_{carb}$ and $\delta^{18}O_{carb}$ values from the lysimeter soils fell between those of geological reference materials, namely sand, limestone used in construction, and concrete aggregates derived from demolition waste (Fig. 4). Overall, $\delta^{18}O_{carb}$ gave a more negative value in the lysimeter BC while the discrepancy in $\delta^{13}C_{carb}$ between two lysimeters was negligible, and $\delta^{13}C_{carb}$ exhibited a strong positive correlation with $\delta^{18}O_{carb}$ ($p < 0.001$, Table S8). Changes in $\delta^{13}C_{carb}$ over time indicated that, like the TOC pool, TIC is dynamic in its composition. Similarly, the $\delta^{18}O_{carb}$ value of topsoil carbonates increased from −12.45 to −9.27‰ in lysimeter BC, and varied little (−9.06 to −9.22‰) in lysimeter WP ($p = 0.007$, Table S7).

3.2. CO_2-carbon flux and concentrations at four soil depths

CO_2–C flux from the topsoil varied greatly, irrespective of measurement method (Table S10). The manually installed tubes method gave an average topsoil flux value for the lysimeter BC of 0.086 ± 0.014 g $m^{-2}{\cdot}h^{-1}$, higher than WP (0.044 ± 0.007 g $m^{-2}{\cdot}h^{-1}$), but without statistical significance. The CO_2–C emission rate of topsoil in WP was initially faster than BC after two months of monitoring; however, later, the CO_2–C flux in both lysimeters was highly variable. The CO_2–C emission rate derived from the manually installed-tubes measurements at 10 cm soil depth recorded a peak value of 0.315 ± 0.008 g $m^{-2}{\cdot}h^{-1}$ in the late spring in 2019 in BC, and peaked at 0.345 ± 0.007 g $m^{-2}{\cdot}h^{-1}$ in July 2019 in WP. Similar trends were reported by the portable CO_2 gas analyzer: the peak CO_2–C emission rate was 0.064 ± 0.057 g $m^{-2}{\cdot}h^{-1}$ in March 2019 in BC, and 0.127 ± 0.009 g $m^{-2}{\cdot}h^{-1}$ in July 2019 in WP, respectively. Additionally, the mean CO_2–C concentrations recorded during the entire gas measurement period at 10 cm, 50 cm, 75 cm depth in the BC lysimeter were 5.35–14.6% greater while this value in the 35 cm soil depth of WP was 0.37% larger, but without any statistical significance, and with great variability indicated by the error bars (Fig. 5; Table S7).

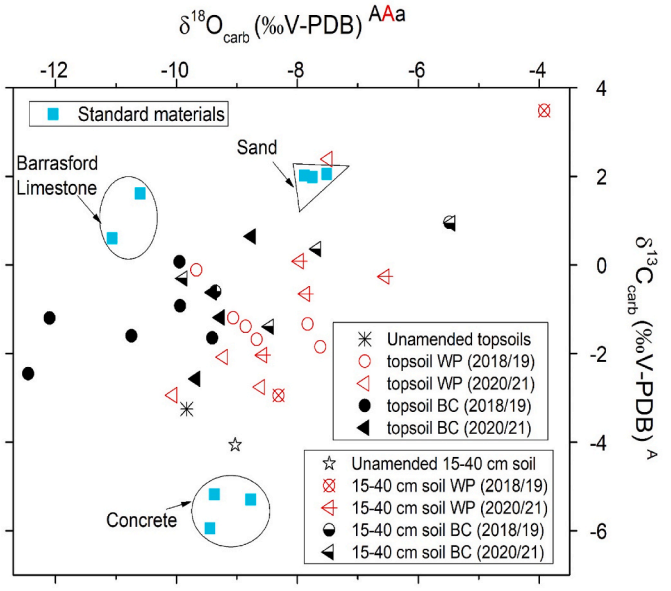


Fig. 4. Measured carbon isotope ($\delta^{13}C_{carb}$) and oxygen isotope ($\delta^{18}O_{carb}$) of carbonate for materials associated with construction (sand, Carboniferous limestone (Barrasford Quarry), and crushed concrete), unamended and amended soil samples from the two lysimeters. BC: lysimeter with wheat straw biochar; WP: lysimeter with wheat straw biomass pellets. The superscript capital letter in black besides the label on each axis indicates significant differences between 2 soil depths in BC ($p \leq 0.05$); the superscript capital letter in red besides the $\delta^{18}O_{carb}$ indicates significant differences between 2 soil depths in WP ($p \leq 0.05$); The superscript lowercase letter in black beside the $\delta^{18}O_{carb}$ indicates significant differences between 2 treatments at $p \leq 0.05$ level. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

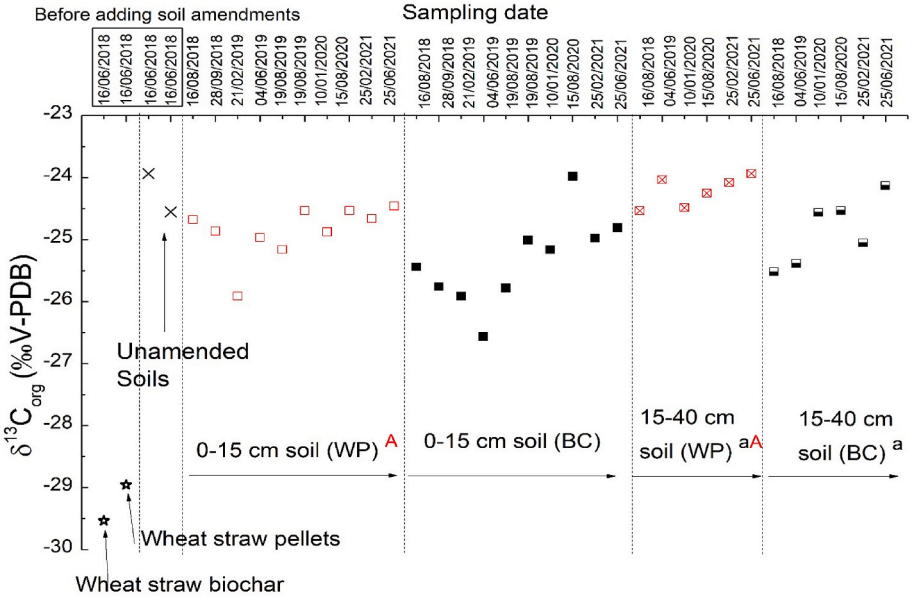


Fig. 3. Measured $\delta^{13}C$ (‰V-PDB) for the organic matter ($\delta^{13}C_{org}$) in the lysimeter soils. BC: lysimeter with wheat straw biochar; WP: lysimeter with wheat straw biomass pellets. The items with same superscript lowercase letters indicate significant differences between 2 treatments at $p \leq 0.05$ level; the items with same superscript capital letters in red indicate significant differences between two soil depths in WP ($p \leq 0.05$ level). (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

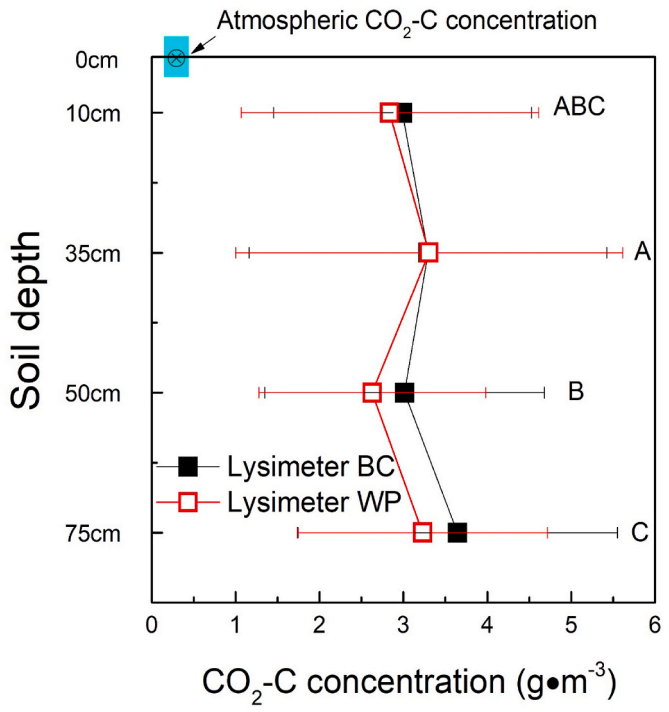


Fig. 5. CO_2–C concentrations ($g \cdot m^{-3}$, Mean ± SD) at specific soil depths in two lysimeters (Aug 2018–Feb 2020). BC: lysimeter with wheat straw biochar; WP: lysimeter with wheat straw biomass pellets. SD: standard deviations are calculated from three replicates per treatment. Same capital letters in black, beside the SD line, indicate significant differences among various soil depths in BC ($p \leq 0.05$ level). There were no significant differences between treatments, and no significant differences among different soil depths in WP.

3.3. Dissolved carbon in leachates

The DTC concentration in leachate from both lysimeters was the highest immediately following the topsoil amendment (Fig. 6), reduced to around 100 mg L^{-1} by October 2018, and then varied within the range of 70–110 mg L^{-1}. In general, leachate from the lysimeter BC had a higher DTC in the first months after the amendment, but afterwards the DTC from the lysimeter WP became higher. On average, the DTC in leachate from lysimeters BC and WP corresponded to 93.31 ± 10.32 mg L^{-1} and 92.13 ± 8.75 mg L^{-1}, respectively, without significant difference ($p = 0.60$, Table S7). But the mean DOC from lysimeter WP (17.90 ± 5.01 mg L^{-1}) was significantly greater than BC (10.49 ± 5.03 mg L^{-1}) ($p < 0.001$, Table S7). Additionally, pH and alkalinity in two treatments decreased with increasing observation time. Leachate from lysimeter BC presented a higher pH and alkalinity, although the statistical correlation only existed in pH ($p < 0.001$, Table S7, Fig. 6).

3.4. Carbon in clippings of above-ground vegetation biomass

The cumulative fresh wet vegetation biomass yield in the lysimeter BC accounted for 73.7% of the yield in the lysimeter WP in the first year, 80.8% in the second year, and 92.9% in the third year following the amendments, respectively (Table 1). Over 3 years, the average carbon of the aboveground vegetation was 0.208 kg m^{-2} in the lysimeter BC and 0.251 kg m^{-2} in WP, without a statistical relationship ($p = 0.488$, Table S7).

3.5. Carbon mass balance for topsoil

The estimated C loss from the lysimeters via leachate over a 36-month period accounted for just above 5% of the C initially added as biochar, or as biomass pellets, respectively (Table S11). Above-ground biomass C produced through plant photosynthesis over 3 years was 6.27% in the lysimeter BC, and 7.57% in WP, compared to the C added as soil amendments. The percentage of C loss from CO_2 emissions relative to the total C input in the two environments could not be ascertained due to high variability in the experimental measurements. However, based on a carbon mass balance it is estimated that the total amount of CO_2–C emitted from the lysimeter WP was over eight times greater than from BC. In other words, the estimated percentage of C loss caused by soil respiration was 13.3% of the initial carbon amendment in the lysimeter BC, while this value is 113% in WP (Table S11).

3.6. Soil water content, temperature, and bulk electrical conductivity

Fig. S1 displays the mean of water content (cm^3/cm^3), soil

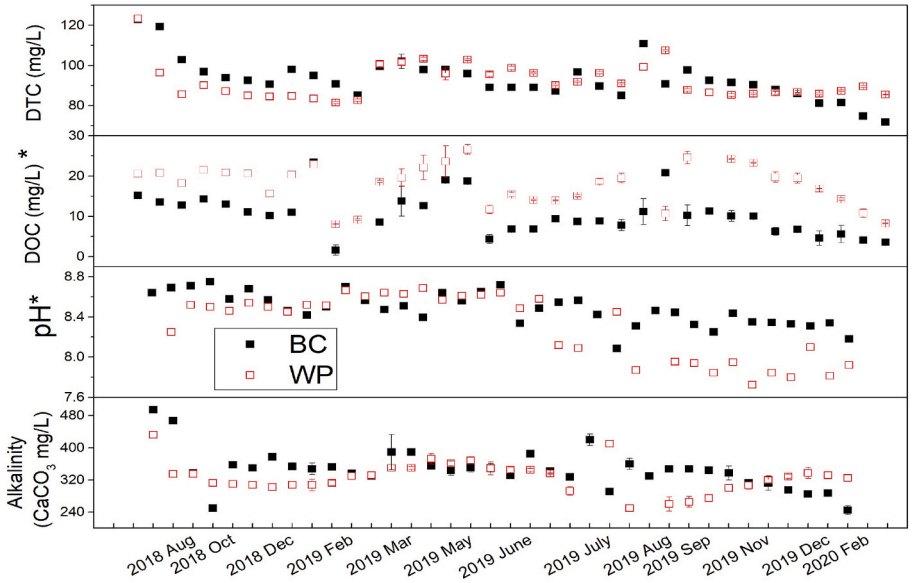


Fig. 6. Dissolved total carbon (DTC, Mean ± SD), dissolved organic carbon (DOC, Mean ± SD), pH, and alkalinity ($CaCO_3$, $mg \cdot L^{-1}$) in the leachate of lysimeters (Aug 2018–Feb 2020). BC: lysimeter with wheat straw biochar; WP: lysimeter with wheat straw biochar pellets. SD refers to standard deviation of replicates. * besides the parameters on the x axis indicates significant differences between 2 treatments ($p \leq 0.05$ level).

Table 1
Carbon content of above-ground biomass in the lysimeters from the annual collection of clippings. BC: lysimeter with wheat straw biochar; WP: lysimeter with wheat straw biomass pellets. No significant difference was found for the above ground biomass carbon.

	Cutting date	Weight of fresh clippings (kg)	Water content (%)	Dry mass of clippings (%)	Weight of dry clippings (kg)	Carbon content of the clippings (kg)	Above-ground biomass carbon storage ($kg \cdot m^{-2}$)
Lysimeter BC	June 11, 2019	7.04	53.0	47.0	3.31	1.57	0.175
	August 15, 2020	10.5	49.2	50.9	5.34	2.54	0.282
	June 17, 2021	7.90	60.0	40.0	3.16	1.50	0.167
Lysimeter WP	June 11, 2019	9.55	61.5	38.5	3.68	1.75	0.194
	August 15, 2020	13.0	51.3	48.7	6.34	3.01	0.334
	June 17, 2021	8.50	50.0	50.0	4.25	2.02	0.224

temperature (°C) and bulk electrical conductivity (mS/m) from sensors placed at 10, 50 and 80 cm soil depth in the lysimeters over 3 years (Fig. 1a). At 10 and 50 cm soil depth, the water content was smaller in the lysimeter BC than WP ($p < 0.001$, Table S7). For soil temperature, the significant difference between the lysimeters only occurred at 50 cm where the average was 2.45% higher in the lysimeter WP. The bulk electrical conductivity at 10 cm was significantly higher in lysimeter WP ($p < 0.001$, Table S7), while this value at 50 cm was significantly higher in lysimeter BC ($p = 0.001$, Table S7). The mean recorded electrical conductivity of topsoil, over 36 months, was 3.92 ± 2.28 mS/m in lysimeter BC and 12.62 ± 11.42 mS/m in lysimeter WP, and significantly increased with soil depth in lysimeter BC (Fig. S1; Table S8).

3.7. Infiltration rates

Infiltration was highly variable, and the average rate in the lysimeter WP (2.34 cm min^{-1}) was higher than that of BC (1.34 cm min^{-1}), but without significant difference (Fig. S2).

3.8. Carbon off-setting potential

If the soil carbon augmentation of the BC over the WP treatment obtained by the end of the three-year period is representative for the longer-term carbon sequestration benefit of applying pyrolyzed instead of unaltered biomass to soil, then the BC application in urban soils could capture and store carbon at around 2.91 ± 2.48 kg m^{-2}, or approximately 30 tonnes·ha^{-1}. Urban land represents 8% of land use in UK (1.8×10^6 ha) (Office for National Statistics, 2019), and its development typically involves earth works which provide an opportunity for embedding carbon-rich amendments. For example, if a biochar amendment as discussed above were applied to the entire blue-green space (1 ha) in the Newcastle Helix site master plan, it would store a total of 30 ± 25 tonnes of carbon, and if conducted over the whole urban green space of Newcastle University, a total of 727 tonnes carbon could be stored. The latter value accounts for 11% of the total stage 1 and 2 CO_2 equivalent-C emissions of Newcastle University (2021) in the academic year 2019/20 (6406 tonnes of CO_2 equivalent-C).

3.9. Stakeholder's perspectives on carbon off-setting strategies

Interviews with key stakeholders at Newcastle University, Newcastle upon Tyne, UK, revealed that several challenges exist in implementing carbon sequestration using biochar produced from local crop residues as soil amendment, from the selection of biochar sources to biochar production and application (Table S12). Firstly, insufficient crop residues are left at the university-managed research farms, and in any case, such residues should be incorporated in the agricultural land to recycle the nutrients into the soil. Thus, utilizing farm residues for producing biochar is not attractive from a farm management perspective (Haefele et al., 2011). On the other hand, the Estate Manager recognized biochar's desirable function of carbon storage which could benefit the institutional target to achieve net-zero carbon by 2030. But they would be concerned about practical problems in terms of dust, storage, haulage, and other factors related to the biochar application. Finally, the leader of the Sustainability Team had questions about the embodied carbon of biochar, the final appearance of amended soil, and how often the biochar application would need repeating (Table S12).

4. Discussion

4.1. Effects of the soil amendments on soil carbon

Wheat straw biochar amendment lastingly improved urban topsoil carbon after three years, while an equivalent amount of carbon in the form of wheat straw biomass could not increase topsoil carbon lastingly (Fig. 2), which aligns with previous reports (Carmo et al., 2016; Huang et al., 2018). With a biochar application rate equivalent to 48 tonnes·ha^{-1}, the TC, TOC and TIC of the urban topsoil in the present study was enhanced by 29.7%, 40.0% and 20.7%, respectively, after 3 years. Similarly, previous studies have shown that soil amendment with biochar enhances soil carbon contents (Carmo et al., 2016; Somerville et al., 2020). After six years, soil carbon increased from 17% to 41% in forest soil for biochar application rates of 2.5 and 25 tonnes·ha^{-1} across the western USA (Sarauer et al., 2019). There was a 3.17% increase of soil TC in biochar-treated arable soil after 2 years with an application rate of 11.5 tonnes·ha^{-1}, compared with untreated soil (Saletnik et al., 2018). In our experiment, the increase of topsoil TOC in the lysimeter BC (40.0%) is significantly greater than the lysimeter WP (8.70%). An eight-year field study of straw and straw-derived biochar amended soils also reported a higher TOC for the biochar application (Sun et al., 2022). Sun et al. (2022), in line with Wang et al. (2021b), attributed this observation to the high alkyl C and aromatic C content of biochar which contributes to soil humic content. Meanwhile, Sun et al. (2022) and Huang et al. (2018) reported that biochar application enhances the proportion of soil macroaggregates, which provides a stable environment for TOC accumulation. By contrast, Yang et al. (2017) observed after three years a higher increase of soil TOC for maize stover application (25%) compared with maize-stover biochar treatment (19%). Higher topsoil carbon in the lysimeter BC confirms that pyrolysis of wheat straw biomass reduces its biodegradability, and hence C mineralization in the soil environment (Zhu et al., 2017). No statistical difference in C contents in the subsurface soil was found between the two lysimeters, which indicates that the biochar amendment was retained in the topsoil layer (Table S7).

4.2. Effects of the soil amendments on soil carbon isotope data

Carbon stable isotope data can indicate the origin of carbon in soil

organic matter and pedogenic carbonate, and enable calculation of the carbon contribution from parent materials and turnover rates (Wang et al., 2015; Dong et al., 2019). $\delta^{13}C$ values for C3 and C4 plants differ, because of differing fractionation of ^{13}C for the two dominant photosynthetic pathways. The $\delta^{13}C$ value for C3 plants, like wheat, is around −27‰ V-PDB, which is similar to our data (wheat straw pellets: −28.95‰; wheat straw biochar: −29.53‰), whereas $\delta^{13}C$ of C4 plants (e.g. maize, sugarcane) is more positive (−13‰) (Christensen et al., 2011; Wang et al., 2015). In addition, $\delta^{13}C$ from atmospheric CO_2 and pedogenic carbonate is less depleted. Cycling carbon from different input sources such as vegetation and litter alters the $\delta^{13}C$ in the soils during cultivation (Christensen et al., 2011). The more reactive carbon pool of wheat straw pellets might become less prominent in the soil $\delta^{13}C$ signal over time, while more stable C is retained in the biochar, all of which drive the distinct changes we observed for $\delta^{13}C$ in wheat straw and biochar amended soils over a period of 3 years. As expected, the original $\delta^{13}C_{org}$ in urban landscaping soil decreased with the amendments because both wheat straw and its biochar present a more negative $\delta^{13}C_{org}$ value (Menichetti et al., 2014; Song et al., 2019; Wang et al., 2015). In both lysimeters, $\delta^{13}C_{org}$ values for amended soil samples remained more negative than the unamended soil, and apart from one outlier (August 15, 2020), this was more obvious in the lysimeter BC ($p = 0.092$, Table S7&S9). Similarly, in a shorter experiment of 180 days, applying cotton straw and its biochar into cropland soils resulted in a more depleted $\delta^{13}C_{org}$ in cotton straw biochar amended soils (Song et al., 2019). The greater stability of the depleted $\delta^{13}C_{org}$ signal in the lysimeter BC shows that a longer lasting TOC augmentation can be achieved with the BC amendment. This trend is consistent with other long-term bare fallow studies in five European cities reported by Menichetti et al. (2014). The experimental duration plays a crucial role in the performance assessment of carbon sequestration with soil amendments as organic inputs from additives will cease at one time point, as well as their potential priming impacts on depressing the microbial activity and metabolism (Menichetti et al., 2014).

The carbon isotope analysis of carbonate can help recognize soil TIC changes. In comparison to the soil collected before treatment, $\delta^{13}C_{carb}$ became markedly more enriched in the topsoil of both lysimeters, but this shift was less pronounced in the WP as compared to the BC lysimeter. The low $\delta^{13}C_{carb}$ of the unamended soil could be due to its content of demolition wastes such as concrete, in which carbonate minerals generated through carbonation post-demolition typically have very low $\delta^{13}C_{carb}$ of −18 to −32‰ (Jorat et al., 2020). $\delta^{13}C_{carb}$ fluctuated around −5.5‰ for Technosol soils in a separate experiment at Cockle Park Farm of Newcastle University (Fig. 4) (Son et al., 2021). A spread in the $\delta^{18}O_{carb}$ values for the carbonates (Fig. 4) can be interpreted as re-equilibration with rain water (Li et al., 2020). There was initially a notable reduction in the TIC content of the topsoil in both lysimeters (Fig. 2). Such a large loss of CO_2 from carbonate dissolution may result from the stimulated activity of bacteria following soil amendment (Sheng et al., 2016). In our study, the soil TIC content was then gradually replenished over time, and more so in the BC as compared to the WP lysimeter (Fig. 2). The newly formed TIC had more enriched $\delta^{13}C_{carb}$ values than the unamended soil TIC, a portion of which was lost following the amendments. Lesser formation of fresh soil TIC over time may then explain why the $\delta^{13}C_{carb}$ in the WP lysimeter remained below that of the BC lysimeter. Furthermore, some of the organic carbon from the amendments with a low $\delta^{13}C_{org}$ may have been mineralised and become part of the $\delta^{13}C_{carb}$ of newly formed carbonates. This probably occurred to a greater extent for the less stable wheat straw pellets, also resulting in a more negative $\delta^{13}C_{carb}$ signal for the topsoil in the WP as compared to the BC lysimeter. Greater TIC formation in the BC lysimeter would be facilitated by 1) the supplement of Ca^{2+} from ash content in biochar (Dong et al., 2019); 2) a higher pH of the biochar amended soil environment; 3) lower electrical conductivity (Khormali et al., 2020); and 4) decreased soil water content (Khormali et al., 2020). In relation to factors 3 and 4, our measurements show that at 10 cm depth, soil electrical conductivity and water content of lysimeter WP was approximately three times and two times greater than BC, respectively. Infiltration rate is also relevant for the dissolution and formation of soil carbonates (Huth et al., 2019). The average infiltration rate for the lysimeter WP in our study is 74.8% faster than for BC, which may have contributed to enhanced weathering of carbonate and the loss of soil TIC in the lysimeter WP (Fig. 2).

4.3. Effects of the soil amendments on C loss via leachate

Adding biochar into soil may reduce the leaching of carbon and nutrients (Iqbal et al., 2015; Liu et al., 2016). As we anticipated, the dissolved organic carbon (DOC) concentration was significantly lower in the lysimeter BC (Zhu et al., 2017; Sun et al., 2022). The DOC change in soils is attributed to the partial decomposition of TOC into leachable fractions such as fulvic acids, while dissolved inorganic carbon (DIC) is driven by the equilibrium of dissolved bicarbonate and carbonate, versus solid $CaCO_3$ (Duffy and Nikolaidis, 2015). On the one hand, a significantly lower DOC in the lysimeter BC might be the result of strong absorption capacity of biochar for organic molecules (Zhu et al., 2017); on the other hand, a higher DOC found in the lysimeter WP could also reflect the wheat straw pellets being a more decomposable substrate than the biochar, and in the longer term also the enhanced vegetation yield of the WP lysimeter. Small molecules such as polysaccharides or peptides are released during straw weathering and form DOC (Sun et al., 2022). Our data substantiate the short-term trends observed over 35-days when sandy loam soil was amended by maize straw or its biochar (Zhu et al., 2017). Zhu et al. (2017) ascribed the greater DOC in leachate of straw treatment to the higher decomposition of straw, and stated that less DOC leaching in the biochar treatment could additionally be caused by enhanced sorption. In contrast, DOC leaching from compost mixed with forest slash biochar was higher than from a compost-only system, but without a statistical significance (Iqbal et al., 2015). Also, Liu et al. (2016) claimed that labile biochar carbon increases DOC in soils amended with straw biochar. In our study, DOC was a lesser contributor to the DTC in leachate than DIC. For DTC, amendment effects were less clear cut, with greater DTC leaching initially from the BC as compared to the WP lysimeter, and an inversion of this trend towards the end of the observations. Similar observations for alkalinity corroborate these findings, since alkalinity typically reflects the DIC content of leachate (i.e. carbonate and bicarbonate). Overall, C loss via leachates over the 3-year observation was in any case only about 5 percent of the C added to the topsoil with either type of amendment (Table S11).

4.4. Effects of the soil amendments on above ground vegetation biomass

In agriculture, applying biomass residues or biochar to soil is a feasible option for improving crop productivity (Huang et al., 2018; Lorenz and Lal, 2014). We observed a higher vegetation productivity in soils amended with wheat straw pellets, as compared to their biochar, during three consecutive years (Table 1), so our second hypothesis for biochar benefits is rejected. According to Carmo et al. (2016), an increase in crop productivity was positively affected by soil electrical conductivity, being in line with our findings (Fig. S1). We embedded an equal amount of organic C into the two lysimeters, while the total amount of additives was different, and thus the proportions of salts and ions may vary accordingly. The higher amount of wheat straw pellets potentially brings more nutrients, which may explain the greater dried grass biomass obtained in the lysimeter WP (Table 1) (Carmo et al., 2016). Similarly, this happened in rape yield when applying rape straw and its biochar, respectively (Huang et al., 2018). High pyrolysis temperature decreases for example the nitrogen content of biochar versus feedstock materials (Chatterjee et al., 2020), and reduced nitrogen availability may be the reason for the lesser vegetation growth in the lysimeter BC versus WP (Table 1). Also, Wang et al. (2021b) explained

that the presence of straw, but not biochar, stimulates soil enzymatic activities, which benefits soil fertility. In our study, the plant productivity in both treatments didn't show a consecutive growth over 3 years, which is consistent with a multi-year and multi-location biochar study on crop yields (Laird et al., 2017). A yield increase in the short-term following organic soil amendments has been attributed to the enhanced fertilization, water-holding capacity, and increased microbial activity (Liu et al., 2016; Saletnik et al., 2018; Carmo et al., 2016). But Major et al. (2010) suggested that such benefits will reduce over time due to leaching of nutrients.

4.5. Effects of the soil amendments on CO_2 emissions

In our research, the monthly CO_2–C fluxes from topsoil in the both lysimeters were highly variable (Table S10). Consequently, for the measured CO_2–C fluxes, the absence of statistically significant differences among 2 systems might be first contributed to insufficient resolution in the data in both space and time (Sarauer et al., 2019). A more apparent relationship between different amendment practices with an important difference tends to happen in short-term research (Ameloot et al., 2013; Wang et al., 2014). For our long-term study, more regular sampling across multiple locations on the surface of each lysimeter would have exceeded the project's financial and human resources. The initial CO_2–C flux (August 2018) under WP treatment is 2.71 times higher than BC treatment (Table S10). Similarly, the CO_2 flux in maize-amended silty loam soil was approximately 3 times that in the biochar amendment for the early incubation stage (Xue et al., 2022); also, the 3–6 times larger respiration rates in three different soils were obtained at the start with straw additions compared to adding biochar (Zhu et al., 2017). These observations indicate more substrate decomposition by microbes in straw-amended systems while biochar has a greater stability due to a high proportion of aromatics and alkenes compounds (Zhu et al., 2017; Wang et al., 2021b), in agreement with Duan et al. (2020) whose straw and biochar amendments were in different sizes. Due to the analytical uncertainty we used a mass balance approach to estimate the overall CO_2–C losses from each lysimeter over the 3 year observation period. Accordingly, the total estimated CO_2–C loss caused by soil respiration is 3.97 kg in lysimeter BC and 33.8 kg in WP. This outcome was anticipated because biochar is associated with lower carbon emissions due to its lower content of labile C (Yang et al., 2017).

4.6. Carbon sequestration opportunity during urban site redevelopments

The redevelopment of urban areas as part of blue-green city planning offers opportunity for carbon off-setting measures that contribute towards net-zero emission targets (Edmondson et al., 2015; Renforth et al., 2011; Scharenbroch et al., 2013). Contrary to our secondary hypotheses, we found that wheat straw amendment outperformed biochar amendment in terms of vegetation growth and infiltration rates in an urban landscaping soil. However, the biochar amendment was beneficial for sequestering carbon not only via TOC enhancement, but also by stimulating TIC formation. For carbon offsetting, biochar deployment into urban soils could thus be a credible policy. Over the whole green space of Newcastle University (25 ha), an estimated 11% of CO_2–C emissions of the University in the academic year 2019–2020 could be sequestered. This calculation also shows that such amendments would need to be used repeatedly or at a very large scale to be relevant for off-setting the annual institutional carbon emissions at current rates. Our previous work found that if Newcastle University transfers all of its managed farmland (805 ha) to coniferous woodland, 50% of its CO_2–C institutional emissions could be off-set by terrestrial carbon sequestration for a period of 40 years (Wang et al., 2021a). However, in the context of blue-green infrastructure, additional benefits might strengthen the case for soil biochar amendments, because biochar can facilitate the water cleansing process and contaminant removal in stormwater management systems (Borah, 2020; Karhu et al., 2011; Renforth et al., 2011). For example, biochar can help ameliorate *Escherichia coli* loads from stormwater owing to its highly porous surface area as well as a strong attachment with hydrophobic attraction to bacteria (Mohanty et al., 2014). Additionally, the effects of soil texture should be considered in future studies, as different organic amendments will influence the size distribution of soil macroaggregates and consequently carbon sequestration (Sun et al., 2022). For instance, the average cumulative CO_2 flux in the top 10 cm layer of sandy loam soil under straw treatment over two years was significantly lower than in the original silty clay loam (Zhang et al., 2021).

4.7. Stakeholder perspectives of biochar application for carbon-offsetting

Stakeholders in Newcastle University were interested in biochar application for carbon sequestration and urban runoff management, in line with other interviews in the UK (Shackley et al., 2011), USA (Krull, 2018), Finland (Salo, 2018) and Sweden (Pensulo, 2012). But biochar application opportunities and benefits need to be advertised, since 7 of 20 gardeners interviewed in the USA hadn't heard of biochar before their interview (Krull, 2018). Like the stakeholders at Newcastle University, respondents across the UK agreed that one of the main concerns for biochar applications is conflict with alternative land use (Shackley et al., 2011). Rural and urban stakeholders in the UK, Sweden, and USA appreciate the adverse effect of climate change on agriculture and cities, but anticipate problems for biochar production and usage such as how to source feedstocks, biochar transportation and deployment, and the lack of funding (Krull, 2018; Pensulo, 2012). Governments could encourage biochar demonstrator projects with subsidies so that stakeholders could experience the biochar application and observe its benefits (Pensulo, 2012; Shackley et al., 2011). Also, for stabilizing the biochar price in an affordable range, financial investments from municipalities would be a key point (Salo, 2018).

5. Conclusions

We applied wheat straw pellets and their biochar produced at 700 °C in a landscaping topsoil to compare the respective amendment benefits for carbon sequestration. The carbon mass balance over three years illustrated that i) the sum of leaching C losses were the same for the two amendments (1.57 kg), ii) the gain from vegetation C fixation was lower under biochar (1.87 kg) as compared to wheat straw practice (2.26 kg), while iii) the gaseous C loss from the wheat straw lysimeter (33.8 kg) was over 8 times larger than from the biochar lysimeter (3.97 kg). These trends were substantiated by the changes observed in soil carbon isotope ratios. Overall, the net carbon accumulation in the biochar-amended topsoil was 2.91 $\pm$ 2.48 kg m^{-2} after 3 years, while carbon loss happened in the wheat straw-amended topsoil (0.36 $\pm$ 1.92 kg m^{-2}). Consequently, 1 ha of blue-green space could sequester around 30 tonnes carbon under wheat straw biochar application. The limited availability of urban land for the implementation of carbon storage is a major challenge when trying to off-set institutional or even city-scale carbon emissions. Furthermore, the availability of crop residues for biochar production, transportation and implementation issues were concerns voiced by potential stakeholders in such off-setting schemes. Future work should i) quantify the net environmental impacts of biochar production and application, ii) examine how soil aggregate-associated carbon and organic functional groups are altered by various amendments, iii) consider how emissions of other greenhouse gases like methane and nitrous oxide are affected, and iv) quantify potential secondary benefits such as water treatment. Such research will enable a more complete appraisal of soil amendment strategies over their whole life cycle.

Contributions

Jiaqian Wang: conceptualization, investigation, methodology, visualization, data curation, software, writing-original draft. David Werner: visualization, methodology, supervision, writing—reviewing and editing. David Manning: supervision, software, writing—reviewing and editing. Ross Stirling: lysimeter establishment, writing-reviewing and editing; Elisa Lopez-Capel: lysimeter establishment, writing-reviewing and editing.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.jenvman.2022.116639.

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